

AE-570: Synthesis of [Ru(CO)(OH)₂(PNP*t*Bu)] using Ag₂O, small batch

Date: 2025-10-01

Tags: Dihydroxo Ag₂O PNP
P(*t*Bu)N(py)P(*t*Bu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

Category: Two-photon

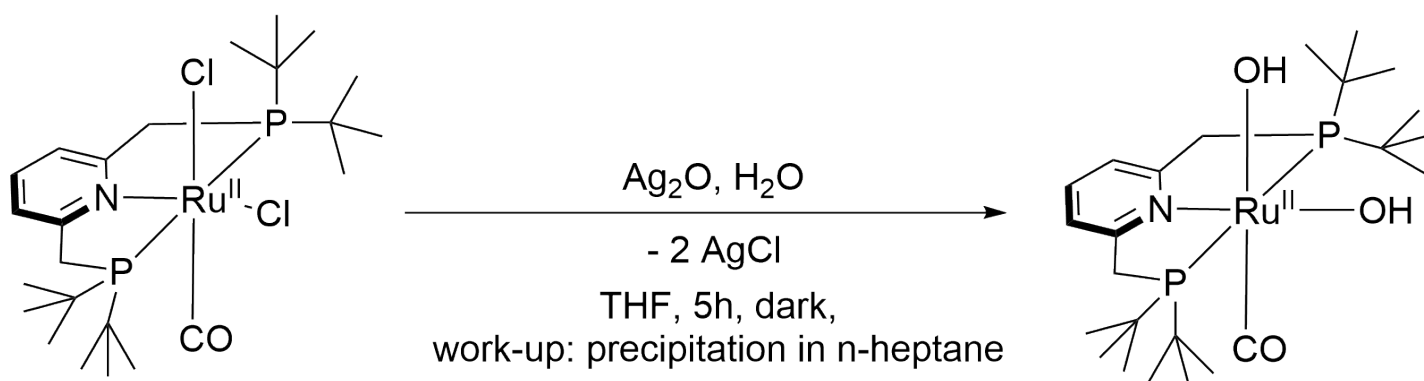
Status: Done

Created by: Alexander Eith

Objective

Resynthesis of [Ru(CO)(OH)₂(PNP)] according to know procedure on a smaller scale, since that was the remaining dichloro

Reaction scheme/sample structure



ChemDraw file: [AE-570.cdxml](#)

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-382: Synthesis of [Ru(CO)(OH) ₂ (PNP <i>t</i> Bu)] using Ag ₂ O, 5 h Two-photon - AE-456: Synthesis of [Ru(CO)(OH) ₂ (PNP <i>t</i> Bu)] using Ag ₂ O, 5 h Two-photon - AE-460: Synthesis of 17O labeled [Ru(CO)(OH) ₂ (PNP <i>t</i> Bu)]

Reagents

Name	CAS-Number/Experiment-Number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume _{theo} [ml]	Volume _{exp} [ml]	Density [g/mL]
[RuCl ₂ (CO)(PNP <i>t</i> Bu)]		0.084	1.00	50	50	595.53	/	/	/
Ag ₂ O	20667-12-3	0.516	6.14	119.47	119.90	231.74	/	/	/
H ₂ O	Prep work - AE-549: Degassing of Milli-Q water	19.7	234	354	/	18.02	0.355	0.35	0.998

THF	Prep work - AE-396: Degassing of THF	/	/	/	/	/	2.55	2.5	/
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Work-up Reagents

Name	CAS Number / Expermient-Number	Inventory number	Mass _{exp} [g]	Volume [ml]	Concentration [M]
THF	Prep work - AE-396: Degassing of THF	/	/	5	/
n-heptane	AE-315: Degassing of n-Heptane	/	/	10 + 2.1	/

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Due to the instability of Ag₂O and Dihydroxo to the light, the procedure was done mostly in the dark, by either wrapping the flasks containing either or both in Aluminium foil or by tuning off the light in the lab and fumehood

Date	Time	Step	Observation	Picture
30.09		A 25 mL brown glas Schlenk flask was prepared transferrered into Equipment - Glovebox CEEC I K004	/	/
	9:20	[RuCl ₂ (CO)(PNP)] was weighted into the reaction schlenk flask and the flask was closed and removed from the glovebox.	/	/
	9:40	Ag ₂ O was added to the flask in air	/	/
	9:45	The flask was set under argon	/	/
	9:58	THF (2.5 mL) was added to the reaction schlenk flask.	/	/
	10:00	Water (0.35 mL) was added to the reaction schlenk flask	/	/
	10:00 - 15:01	The reaction mixture was stirred for 5 h at 700 rpm at rt in the dark (wrapped with aluminium foil	/	/
	15:01 - 15:08	The reaction mixture was allowed to settle	/	/
	15:08	The mixture was filtered through a PTFE syringe filter (pore-diameter: 0.2 µm) into a prepared 20 mL flask	black suspension violet solution	before filt.jpg after filt.jpg
	15:14	The solvents of the filtrate were removed under reduced pressure. No water bath was used	/	frozen.jpg

	15:17	After the mixture solidified a water bath was added	This procedure avoids bubbling up of the reaction mixture	/
	15:24	The obtained solid was dried under reduced pressure	reddish-brwonish film	dried.jpg
	16:30	The solid was stored in the dark at rt	no change	solid after night.jpg
01.10	11:20	To the solid THF (5 mL) was added und the obtained mixture was shaken strongly for approx. 20 s and than stirred for approx. 5 min at 400 rpm	cloudy suspensnion	/
	11:35	n-heptane (10 mL) was added to the mixture	beige precipitate, violte/pink solution	precipitated in n-heptane.jpg
	11:37	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 100 mL-Schlenk	violet filtrate, beige solid	after filt.jpg
	11:55	The solid was washed with n-heptane (3 * 0.7 mL)	no change	after washing.jpg
	11:57 - 14:50	The solid was dried under reduced pressure	/	/
		The solid was stored in Equipment - Freezer Lab E004 CEEC II at -20 °C in the dark under Argon	/	/
08.10	15:30	The solid was transferred into a 2 mL vial in a 10 mL Schlenk flask	/	/
		During this an aliquote (2.12 mg) was transferred into another 10 mL Schlenk flask	/	/
	15:45	Both flask were set under argon	grayish-beige solid	Prod.jpg
	16:02	From the aliquote n NMR sample was prepared in DMSO-d6 (0.7 mL) according to Protocol - Preparation of NMR Sample	AE-570-prod	in DMSO-d6.jpg AE-570-prod.jpg

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
09.10	10:12	AE-570-prod	NMR 1H, 31P{1H}quant	IAAC 400 MHz I	DMSO-d6	2025-10-09_13-52-59.zip /		2025-10-09_13-52-59.zip_AE-570-PROD_10.nmrium	Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h	Signals as expected. In 1 H, as expected only minor impruties, in 31 P 1 further signal with 2 %

Product characterization

Sample	Mass [mg]	Purity	Mass _{pure} [mg]	Amount [μmol]	Yield [%]	Description	Photo	Storage location
AE-570-prod	14.29	98 %	14.00	25.1	30	grayish-beige solid	Prod.jpg	Equipment - Freezer Lab E004 CEEC II

Result

Linked experiments

- [AE-315: Degassing of n-Heptane](#)

Prep work - [AE-396: Degassing of THF](#)

Prep work - [AE-549: Degassing of Milli-Q water](#)

Two-photon - [AE-382: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

Two-photon - [AE-456: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

Two-photon - [AE-460: Synthesis of ¹⁷⁰labeled \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\]](#)

Linked resources

Equipment - [Freezer Lab E004 CEEC II](#)

Equipment - [Glovebox CEEC I K004](#)

Equipment - [Glovebox CEEC I K004](#)

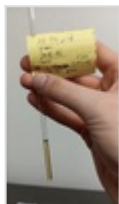
Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

Protocol - [Schlenk Technique](#)

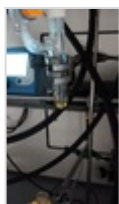
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in DMSO-d6.jpg

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Prod.jpg

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after washing.jpg

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after filt.jpg

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precipitated in n-heptane.jpg

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solid after night.jpg

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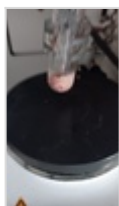
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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3062>